

4.11.38

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Question

Find the area of the region bounded by the lines $y = 2x + 1$, $y = 3x + 1$ and $x = 4$.

Solution

Given three lines are

$$\begin{pmatrix} -2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 1 \implies \mathbf{n}^\top \mathbf{x} = 1 \quad (1)$$

$$\begin{pmatrix} -3 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 1 \implies \mathbf{m}^\top \mathbf{x} = 1 \quad (2)$$

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 4 \implies \mathbf{p}^\top \mathbf{x} = 4 \quad (3)$$

Solution

The three lines form a triangle. The vertices of triangle are obtained by **Intersection of : (1) and (3)**

$$\left(\begin{array}{cc|c} -2 & 1 & 1 \\ 1 & 0 & 4 \end{array} \right) \xrightarrow{R_1 \rightarrow R_1 + 2R_2} \left(\begin{array}{cc|c} 0 & 1 & 9 \\ 1 & 0 & 4 \end{array} \right) \quad (1)$$

From the second row we get $y = 9$ so $x = 4 \implies \mathbf{A} = \begin{pmatrix} 4 \\ 9 \end{pmatrix}$

Intersection of : (2) and (3)

$$\left(\begin{array}{cc|c} -3 & 1 & 1 \\ 1 & 0 & 4 \end{array} \right) \xrightarrow{R_1 \rightarrow R_1 + 3R_2} \left(\begin{array}{cc|c} 0 & 1 & 13 \\ 1 & 0 & 4 \end{array} \right) \quad (2)$$

From the second row we get $y = 13$ so $x = 4 \implies \mathbf{B} = \begin{pmatrix} 4 \\ 13 \end{pmatrix}$

Intersection of : (1) and (2)

$$\left(\begin{array}{cc|c} -2 & 1 & 1 \\ -3 & 1 & 1 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - R_1} \left(\begin{array}{cc|c} -2 & 1 & 1 \\ -1 & 0 & 0 \end{array} \right) \quad (3)$$

From the second row we get $x = 0$ so $y = 1 \implies \mathbf{C} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Solution

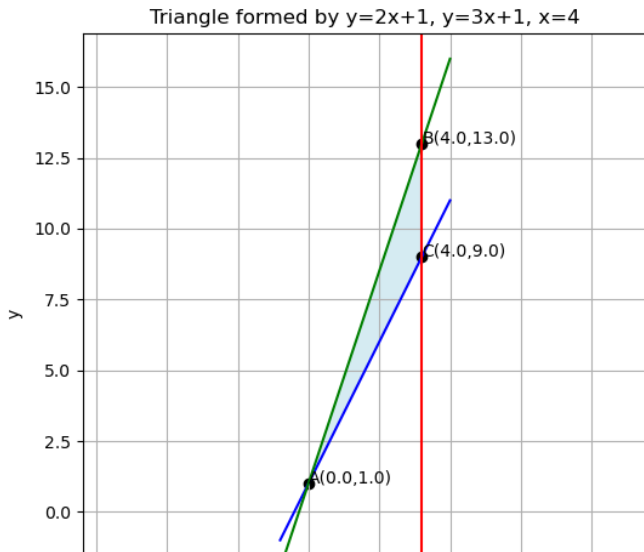
$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} 0 \\ -4 \end{pmatrix}, \quad \mathbf{A} - \mathbf{C} = \begin{pmatrix} 4 \\ 8 \end{pmatrix} \quad (4)$$

$$\|(\mathbf{A} - \mathbf{B}) \times (\mathbf{A} - \mathbf{C})\| = \begin{vmatrix} 0 & -4 \\ 4 & 8 \end{vmatrix} = |0 - (-16)| = 16 \quad (5)$$

$$\text{Area of the triangle} = \frac{1}{2} \|(\mathbf{A} - \mathbf{B}) \times (\mathbf{A} - \mathbf{C})\| = \frac{16}{2} = 8 \quad (6)$$

Area of the triangle formed by the lines $y = 2x + 1$, $y = 3x + 1$ and $x = 4$ is 8 square units.

Graph



```
#include <math.h>

static void intersect(double a1,double b1,double c1,
                    double a2,double b2,double c2,
                    double *x,double *y)
{
    double D = a1*b2 - a2*b1;
    *x = (b1*(-c2) - b2*(-c1)) / D;
    *y = (a2*(-c1) - a1*(-c2)) / D;
}

// Problem: y=2x+1, y=3x+1, x=4
// In ax+by+c=0 form: -2x + y -1 = 0 ; -3x + y -1 = 0 ; 1x + 0y
//                  -4 = 0
double triangle_area_example(void)
```



```
{  
double x1,y1,x2,y2,x3,y3;  
  
// L1 & L2  
intersect(-2, 1, -1, -3, 1, -1, &x1,&y1);  
// L2 & L3  
intersect(-3, 1, -1, 1, 0, -4, &x2,&y2);  
// L3 & L1  
intersect( 1, 0, -4, -2, 1, -1, &x3,&y3);  
  
// Area = 1/2 * |det([B-A, C-A])|  
double ABx = x2 - x1, ABx = y2 - y1;  
double ACx = x3 - x1, ACy = y3 - y1;  
double area = 0.5 * fabs(ABx*ACy - ABx*ACx);  
return area;  
}
```

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
import os

def intersection(a1,b1,c1, a2,b2,c2):
    A = np.array([[a1,b1],[a2,b2]], dtype=float)
    B = np.array([-c1,-c2], dtype=float)
    x, y = np.linalg.solve(A, B)
    return np.array([x, y])

L1 = (-2, 1, -1)
L2 = (-3, 1, -1)
L3 = (1, 0, -4)

A = intersection(*L1, *L2)
B = intersection(*L2, *L3)
C = intersection(*L1, *L3)
V = np.vstack([A,B,C])
```

```
x1,y1 = A; x2,y2 = B; x3,y3 = C
area = 0.5*abs(x1*(y2-y3)+x2*(y3-y1)+x3*(y1-y2))
print(Area of the triangle =, area)
xs = np.linspace(min(V[:,0])-1, max(V[:,0])+1, 400)

plt.figure(figsize=(6,6))
plt.plot(xs, 2*xs + 1, color=blue)
plt.plot(xs, 3*xs + 1, color=green)
plt.axvline(4, color=red)
plt.fill([x1,x2,x3],[y1,y2,y3], color=lightblue, alpha=0.5)
```

```
for name, P in zip([A,B,C], V):
    plt.scatter(P[0], P[1], color=black)
    plt.text(P[0], P[1], f{name}({P[0]:.1f},{P[1]:.1f}))

plt.xlabel(x)
plt.ylabel(y)
plt.title(Triangle formed by  $y=2x+1$ ,  $y=3x+1$ ,  $x=4$ )
plt.grid(True)
plt.axis(equal)

os.makedirs(figs, exist_ok=True)
plt.savefig(figs/triangle.png, dpi=200)
plt.show()
```

```
import ctypes
from ctypes import c_double
import os, sys

# Pick correct lib name
libname = libsimple_triarea.dylib if sys.platform == darwin else
libsimple_triarea.so
lib = ctypes.CDLL(os.path.abspath(libname))

# C function: double triangle_area_example(void)
lib.triangle_area_example.restype = c_double

area = lib.triangle_area_example()
print(fArea of the triangle (y=2x+1, y=3x+1, x=4) = {area})
```